

Exercício 01

$\text{max}(X, Y, Z) :- X \geq Y, Z = X.$

$\text{max}(X, Y, Z) :- X < Y, Z = Y.$

Exercício 02

$\text{max_list}([H], H).$

$\text{max_list}([H|T], M) :-$

$\text{max_list}(T, MT),$

$\text{max}(H, MT, M).$

Exercício 03

$\text{ordered}([]).$

$\text{ordered}([_]).$

$\text{ordered}([X, Y | T]) :-$

$X \leq Y,$

$\text{ordered}([Y | T]).$

Exercício 04

$\text{split}([], [], []).$

$\text{split}([X], [X], []).$

$\text{split}([X, Y | T], [X | T1], [Y | T2]) :-$

$\text{split}(T, T1, T2).$

$\text{merge}([], L, L).$

$\text{merge}(L, [], L).$

$\text{merge}([X | T1], [Y | T2], [X | T]) :-$

$X \leq Y,$

$\text{merge}(T1, [Y | T2], T).$

$\text{merge}([X | T1], [Y | T2], [Y | T]) :-$

$X > Y,$

$\text{merge}([X | T1], T2, T).$

$\text{merge_sort}([], []).$

$\text{merge_sort}([X], [X]).$

$\text{merge_sort}(\text{In}, \text{Out}) :-$

$\text{split}(\text{In}, L1, L2),$

$\text{merge_sort}(L1, \text{SortedL1}),$

$\text{merge_sort}(L2, \text{SortedL2}),$

$\text{merge}(\text{SortedL1}, \text{SortedL2}, \text{Out}).$

Exercício 05

$\text{nocheck}(_, []).$

$\text{nocheck}(X/Y, [X1/Y1 | T]) :-$

$Y \neq Y1,$

$\text{abs}(X - X1) \neq \text{abs}(Y - Y1),$

$\text{nocheck}(X/Y, T).$

$\text{legal}([]).$

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legal([X/Y | T]) :-
    legal(T),
    member(Y, [1,2,3,4,5,6,7,8]), % Para NxN, substitua por uma lista dinâmica.
    nocheck(X/Y, T).

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n_queens(N, X) :-
    findall(I, between(1, N, I), Indices), % Gera uma lista [1, 2, ..., N].
    length(X, N),
    maplist(between(1, N), X), % Garante que cada Y está no intervalo correto.
    maplist(\=(X), Indices), % Configuração inicial.
    legal(X).

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Exercício 6:

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subset_sum([], 0, []).
subset_sum([H | T], Sum, [H | SubL]) :-
    Remaining is Sum - H,
    subset_sum(T, Remaining, SubL).
subset_sum(_ | T, Sum, SubL) :-
    subset_sum(T, Sum, SubL).

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